SUNGAUSS: Physically-Based Sun Relighting for In-the-Wild Outdoor Scenes via Gaussian Splatting

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***Abstract*—We present SUNGAUSS, a method for relighting outdoor scenes captured in uncontrolled conditions under natural sunlight. Built upon 2D Gaussian Splatting (2DGS) with appearance embeddings, we replace the per-image spherical-harmonic environment maps of prior work with an explicit physically-motivated sun lighting model that decomposes illumination into a direct solar term, per-image ambient light, and a shared global sky component represented as low-order spherical harmonics. Our explicit decomposition enables three capabilities absent from the baseline: (i) geometry-based shadow computation via differentiable shadow mapping from an orthographic sun camera, (ii) optional Cook-Torrance physically-based rendering with per-Gaussian roughness and metallic parameters, and (iii) realistic day-cycle relighting by simply sweeping the sun direction through a solar trajectory. We further introduce a learnable per-Gaussian sky distinction parameter supervised by binary sky masks, an adaptive voxel-based densification strategy driven by per-pixel reconstruction loss with high-frequency awareness, and support for both surfel (2DGS) and classical 3D Gaussian rasterization backends. [TODO: Quantitative results on NeRF-OSR benchmark.]**

I. INTRODUCTION

Relighting outdoor scenes from uncontrolled photo collections is a long-standing challenge in computer vision and graphics. Unlike studio environments with known illumination, in-the-wild captures exhibit varying sun position, cloud cover, and atmospheric conditions across images. Recent neural scene representations—particularly Neural Radiance Fields (NeRFs) [1] and Gaussian Splatting [2]—have enabled photorealistic novel-view synthesis, but most methods bake illumination into the scene representation, preventing post-capture relighting.

LumiGauss [4] addresses this for outdoor scenes by combining 2D Gaussian Splatting (2DGS) [3] with per-image appearance embeddings and spherical-harmonic (SH) environment maps. Each image is assigned a learnable SH environment that modulates Gaussian colors, enabling factorization of geometry, albedo, and illumination. However, the per-image SH representation has important limitations for outdoor relighting:

- SH environments are unconstrained and per-image—they do not share structure across images, making it impossible to interpolate to novel, unseen illumination conditions.
- Shadows are encoded only implicitly through per-Gaussian SH feature modulation, which cannot produce sharp cast shadows from geometry.
- There is no explicit notion of the sun as a directional source, precluding physically-grounded relighting such as time-of-day simulation.

We address these limitations with SUNGAUSS, which makes the following contributions:

1) **Explicit directional sun model** (§III-A): A physically-motivated lighting decomposition separating direct sunlight (with Lambert shading and an atmospheric color prior), per-image ambient light, and a shared global sky SH, enabling novel illumination synthesis.
2) **Geometry-based shadow computation** (§III-C): Multiple shadow algorithms—shadow mapping via orthographic sun projection, volumetric ray marching, and voxel-based tracing—applied to the direct lighting term.
3) **Learnable sky-scene distinction** (§III-D): A per-Gaussian continuous parameter supervised by sky masks that controls both shadow casting and shadow receiving.
4) **Adaptive loss-driven densification** (§III-E): A voxel grid that accumulates per-pixel photometric loss in 3D, identifies under-reconstructed regions via high-frequency and hole-filling heuristics, and spawns new Gaussians where needed.
5) **Optional Cook-Torrance PBR** (§III-F): Per-Gaussian roughness and metallic parameters enabling a microfacet BRDF for scenes with specular materials.

II. RELATED WORK

a) Neural Scene Relighting.: NeRF-OSR [5] decomposes outdoor NeRF scenes into albedo and per-image SH lighting, enabling appearance transfer between views. NeRF-W [6] handles appearance variation with per-image latent codes. More recently, methods based on 3D Gaussian Splatting [2] have achieved real-time rendering while maintaining relighting capability.

b) Outdoor Illumination Modeling.: Modeling outdoor illumination typically involves representing the sun as a directional source and the sky as a hemisphere of diffuse light [10], [11]. The color of sunlight varies with elevation due to atmospheric Rayleigh scattering, producing the characteristic warm tones at sunrise and sunset [8], [9]. Our sun color prior draws on this atmospheric model.

c) Shadow Computation.: Shadow mapping [7] renders a depth map from the light's viewpoint and compares fragment depths to determine occlusion. In the Gaussian splatting setting, the unstructured point-based representation requires

adapting shadow mapping to work with splatted depth rather than mesh rasterization. We additionally implement volumetric approaches—ray marching through Gaussian fields and voxel-based shadow tracing—as alternatives for different quality/speed trade-offs.

d) 2D Gaussian Splatting.: 2DGS [3] represents scene geometry as oriented planar splats (surfels), providing well-defined surface normals from the splat orientation. LumiGauss [4] builds on 2DGS, leveraging the surfel normals for hemisphere-based SH evaluation. We inherit this representation and additionally leverage the normals for Lambert shading (N·L) and optional PBR.

e) Adaptive Densification.: The original 3DGS [2] densifies Gaussians based on view-space gradient magnitude, which can miss under-reconstructed regions that happen to have low gradient. Several works have proposed improved densification strategies [13], [14]; our approach operates in 3D voxel space and is driven by photometric loss, providing spatial awareness absent from gradient-only methods.

III. Method

Given a set of posed images $\{I_i\}_{i=1}^N$ of an outdoor scene with known per-image sun directions $\{\mathbf{l}_i\}$ and optional binary sky masks $\{M_i^{\text{sky}}\}$, our goal is to learn a Gaussian scene representation that decomposes geometry, material (albedo), and illumination, enabling relighting under novel sun conditions.

We build on the 2DGS surfel representation [3] with per-image appearance embeddings from LumiGauss [4]. Our method replaces the per-image SH environment with an explicit sun lighting model, adds geometry-based shadow computation, and introduces several training improvements. Figure 1 provides an overview.

A. Explicit Directional Sun Model

We decompose the illumination reaching each Gaussian into three physically-motivated terms:

$$L(\mathbf{x},\mathbf{n}) = \underbrace{C_{\text{sun}}(\theta)\cdot I_i \cdot \max(0,\mathbf{n}\cdot\mathbf{l}_i)}_{\text{direct sunlight}} + \underbrace{\mathbf{c}_i^{\text{amb}}}_{\text{ambient}} + \underbrace{E_{\text{sky}}(\mathbf{n})}_{\text{global sky}} \quad (1)$$

where $\mathbf{n}$ is the surface normal (derived from the surfel quaternion and scale, §III-B), $\mathbf{l}_i$ is the sun direction for image i, and:

a) Sun color prior $C_{sun}(\theta)$.: We model the spectral shift of sunlight with elevation θ using a simplified Rayleigh scattering model. The transmittance per RGB channel is:

$$T_c = \exp\left(-\beta_R \cdot \text{AM}(\theta)\cdot\left(\frac{\lambda_c}{650}\right)^{-4}\cdot s\right) \quad (2)$$

where $\text{AM}(\theta) = 1/\sin(\theta)$ is the air mass, $\lambda_c \in \{650, 550, 450\}$ nm for R, G, B, $\beta_R = 0.05$ is the Rayleigh coefficient, and $s = 8.0$ is a scattering scale. This produces physically plausible warm tones at low elevations and neutral white at zenith.

b) Per-image parameters.: Each image has three learnable parameters:

- $I_i \in \mathbb{R}^+$: sun intensity scalar (initialized to 4.0)
- $\mathbf{k}_i \in [0.5, 2.0]^3$: color correction multiplier on C_{sun}, accommodating cloud cover and atmospheric variation
- $\mathbf{c}_i^{\text{amb}} \in \mathbb{R}_+^3$: ambient/indirect illumination (initialized to 0.15)

The effective sun color is $C_{\text{sun}}(\theta)\cdot\mathbf{k}_i\cdot I_i$.

c) Global sky SH.: We represent the directional sky illumination as degree-1 spherical harmonics $E_{\text{sky}} \in \mathbb{R}^{3\times 4}$, shared across all images. This captures the dominant sky gradient (bright horizon, dark zenith) while remaining low-dimensional enough to constrain the optimization. Evaluated at normal $\mathbf{n}$:

$$E_{\text{sky}}(\mathbf{n}) = \sum_{l=0}^{1}\sum_{m=-l}^{l} c_{lm}\cdot Y_l^m(\mathbf{n}) \quad (3)$$

d) Final per-Gaussian color.: The HDR intensity is gamma-corrected and multiplied by the learned albedo $\mathbf{a}$:

$$\text{RGB} = \mathbf{a}\cdot(L(\mathbf{x},\mathbf{n}))^{1/\gamma}, \quad \gamma = 2.2 \quad (4)$$

B. Surface Normals from Surfel Geometry

Each 2DGS surfel is parameterized by a quaternion $\mathbf{q}$ and anisotropic scales $\mathbf{s}\in\mathbb{R}^3$. The surfel plane normal is the third column of the rotation matrix $R(\mathbf{q})$:

$$\mathbf{n}_w = R(\mathbf{q})_{:,3} \quad (5)$$

We apply a view-dependent sign flip to ensure normals face the camera:

$$\mathbf{n} = \text{sign}(-\mathbf{n}_w\cdot\mathbf{p}_{\text{view}})\cdot\mathbf{n}_w \quad (6)$$

where $\mathbf{p}_{\text{view}}$ is the Gaussian position in camera coordinates. This yields consistent camera-facing normals for Lambert shading.

C. Shadow Computation

Shadows are applied exclusively to the direct sunlight term; ambient and sky illumination remain unaffected. The shadowed intensity for Gaussian j in image i becomes:

$$L_{\text{shad}} = C_{\text{sun}}\cdot I_i\cdot(\mathbf{n}\cdot\mathbf{l}_i)\cdot\sigma_j + \mathbf{c}_i^{\text{amb}} + E_{\text{sky}}(\mathbf{n}) \quad (7)$$

where $\sigma_j \in [0,1]$ is the shadow mask (1 = lit, 0 = shadowed).

We implement three shadow computation methods:

1) Shadow Mapping: Our primary method adapts classical shadow mapping [7] to Gaussian splatting. We construct an orthographic sun camera positioned along $\mathbf{l}_i$ at distance $3\times r_{\text{scene}}$ from the scene center, with frustum half-width $1.5\times r_{\text{scene}}$, where r_{scene} is the maximum distance from any Gaussian to the scene centroid.

We render this view using the surfel rasterizer to obtain a depth map $D_{\text{sun}}\in\mathbb{R}^{R\times R}$ and alpha map α_{sun}. Sky Gaussians are excluded by zeroing their opacity using the learned w_j^{sky} parameter (§III-D).

[TODO: Pipeline figure: show Gaussians → normals → N·L → shadow map → composited image]

Fig. 1: Overview of the SUNGAUSS pipeline. Per-Gaussian normals from the surfel orientation are used for Lambert shading with the sun direction. A shadow map rendered from an orthographic sun camera provides per-Gaussian shadow masks. The shadowed direct light, ambient, and sky SH components are composed and multiplied by learned albedo to produce the final image.

For each Gaussian j with world position $\mathbf{x}_j$, we project into the sun camera's clip space, bilinearly sample the shadow depth map, and compare:

$$\sigma_j = \begin{cases} 0 & \text{if } d_j > D_{\text{sun}}(\mathbf{u}_j) + b \ \wedge \ \alpha_{\text{sun}}(\mathbf{u}_j) > 0.1 \\ 1 & \text{otherwise} \end{cases} \tag{8}$$

where d_j is the projected depth, $\mathbf{u}_j$ is the UV coordinate, and b is a bias term to prevent self-shadowing. Gaussians outside the shadow frustum default to lit.

2) Volumetric Ray Marching: For each Gaussian, we march a ray toward the sun through the scene, accumulating opacity contributions from nearby Gaussians using a spatial hash grid for acceleration. The shadow factor is:

$$\sigma_j = \exp\left(-2 \sum_{k=1}^{K} o_k(\mathbf{x}_j + k\Delta \mathbf{l})\right) \tag{9}$$

where o_k is the maximum Gaussian opacity contribution at step k and K is the number of march steps. This produces naturally soft shadows modulated by Gaussian opacity.

3) Voxel-Based Tracing: The scene is voxelized into a V^3 grid by splatting Gaussian opacity weighted by a Gaussian kernel. Rays are marched from each Gaussian toward the sun through the voxel grid, accumulating opacity per step. This method trades accuracy for speed and is suitable for preview rendering.

D. Learnable Sky-Scene Distinction

Each Gaussian carries a continuous parameter $w_j^{\text{sky}} \in [0, 1]$, implemented as a clamped learnable scalar. This parameter serves a dual role:

- **Shadow casting**: In the shadow map, each Gaussian's opacity is multiplied by w_j^{sky}, so sky Gaussians ($w_j^{\text{sky}} \approx 0$) become transparent to the sun.
- **Shadow receiving**: Sky Gaussians are forced to fully lit ($\sigma_j = 1$) regardless of the shadow computation, preventing the sky from being darkened by shadows.

a) Supervision.: Given binary sky masks M^{sky} (1 = non-sky, 0 = sky), we render each view using w_j^{sky} as the per-Gaussian color and supervise with an L1 loss:

$$\mathcal{L}_{\text{sky}} = \lambda_{\text{sky}} \left\| \text{Render}(w^{\text{sky}}) - M^{\text{sky}} \right\|_1 \tag{10}$$

This is differentiable through the rasterizer and jointly optimized with all other parameters.

E. Adaptive Loss-Driven Densification

Standard gradient-based densification in 3DGS [2] adds Gaussians where the view-space position gradient is large, but can miss regions with consistent but high reconstruction error, or fail to fill holes in the geometry. We propose a complementary voxel-based densification strategy driven directly by the photometric loss.

1) Loss Accumulation in 3D: We maintain a voxel grid $\mathcal{G} \in \mathbb{R}^{G^3}$ (default $G = 48$) spanning the scene's axis-aligned bounding box. At each training iteration, for each pixel with loss $\ell > 0$, we unproject it to 3D using the rendered depth and camera intrinsics, then scatter-add the loss to the corresponding voxel:

$$\mathcal{G}[\text{vox}(\mathbf{x}_{\text{3D}})] \mathrel{+}= \ell(\mathbf{x}_{\text{3D}}) \tag{11}$$

a) High-frequency boosting.: Under-reconstructed textured regions should receive higher densification priority. We compute a Sobel edge magnitude map on the reference image and boost the loss contribution for pixels exceeding the q_{hf}-quantile of edge magnitude:

$$\ell' = \ell \cdot (1 + \beta_{\text{hf}} \cdot \mathbb{1}[\text{Sobel}(\mathbf{x}) > Q_{q_{\text{hf}}}]) \tag{12}$$

where β_{hf} is the boost factor (default 0.75) and q_{hf} is the quantile threshold (default 0.6).

b) Hole filling.: Pixels with zero rendered depth indicate geometric holes. For these pixels, we cast a ray through the scene AABB using slab-method ray-AABB intersection and sample random 3D points along the valid ray segment. Only holes with loss above a configurable quantile are populated, preventing densification in truly empty regions (e.g., sky).

2) Triggering and Budget Allocation: Every T iterations (default 250), we evaluate the accumulated grid:

1) Apply temporal EMA smoothing: $\bar{\mathcal{G}} = \max(\mathcal{G}_{\text{avg}}, \alpha \cdot \text{EMA} + (1 - \alpha) \cdot \mathcal{G}_{\text{avg}})$
2) Select candidate cells: observed, sparse (Gaussian count below threshold), and high-loss (above configurable quantile)
3) Optionally filter by camera-center distance to prevent hallucination in poorly observed regions
4) Allocate a budget B (default 4096 Gaussians) to selected cells, weighted by loss magnitude and sparsity deficit

New Gaussians are initialized at random positions within their cell, with small scale, low opacity (0.12), and albedo set to the scene mean, supporting the subsequent optimization to refine them.

F. Cook-Torrance PBR Rendering

For scenes with specular materials, we optionally augment each Gaussian with roughness $r_j \in [0.04, 1]$ and metallic $m_j \in [0, 1]$ parameters, and replace the Lambert-only shading with a full Cook-Torrance microfacet BRDF:

$$f(\mathbf{l}, \mathbf{v}) = \frac{(1-F)(1-m)\,\mathbf{a}}{\pi} + \frac{D \cdot G \cdot F}{4(\mathbf{n}\cdot\mathbf{v})(\mathbf{n}\cdot\mathbf{l})} \tag{13}$$

where:

$$D = \frac{\alpha^2}{\pi\left((\mathbf{n}\cdot\mathbf{h})^2(\alpha^2-1)+1\right)^2} \quad \text{(GGX NDF)} \tag{14}$$

$$G = G_1(\mathbf{v})\,G_1(\mathbf{l}), \quad G_1(\mathbf{x}) = \frac{\mathbf{n}\cdot\mathbf{x}}{\mathbf{n}\cdot\mathbf{x}(1-k)+k} \quad \text{(Schlick-GGX)} \tag{15}$$

$$F = F_0 + (1-F_0)(1-\mathbf{v}\cdot\mathbf{h})^5 \quad \text{(Schlick)} \tag{16}$$

with $\alpha = r^2$, $k = (r+1)^2/8$, $F_0 = 0.04(1-m) + \mathbf{a}\cdot m$, and $\mathbf{h} = (\mathbf{l}+\mathbf{v})/\|\mathbf{l}+\mathbf{v}\|$.

The indirect (ambient + sky) contribution is modulated by material properties:

$$L_{\text{ind}} = \mathbf{a}(1-m)(\mathbf{c}^{\text{amb}} + E_{\text{sky}}) + F_0 \cdot 0.25 \cdot (\mathbf{c}^{\text{amb}} + E_{\text{sky}}) \tag{17}$$

IV. Training

Training proceeds in three phases to ensure stable optimization:

a) Phase 1: Geometry warmup (iterations 0–T_w).: Only the unshadowed rendering loss is active, using Eq. 1 without shadows. This allows the geometry, albedo, sun parameters, and sky SH to converge to a consistent state before introducing the more complex shadowed loss.

b) Phase 2: Transition (T_w–T_s).: Learning rates for lighting parameters are adjusted. Standard gradient-based densification and pruning remain active.

c) Phase 3: Shadowed training (iterations $> T_s$).: The full shadowed rendering (Eq. 7) becomes the primary photometric loss. Shadow maps are recomputed each iteration. Adaptive densification is activated if enabled.

d) Loss function.: The total loss combines:

$$\mathcal{L} = \lambda_{\text{shad}}\mathcal{L}_{\text{img}}^{\text{shad}} + \lambda_{\text{unshad}}\mathcal{L}_{\text{img}}^{\text{unshad}} + \lambda_{\text{sky}}\mathcal{L}_{\text{sky}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} \tag{18}$$

where $\mathcal{L}_{\text{img}} = (1-\lambda_{\text{ssim}})\mathcal{L}_1 + \lambda_{\text{ssim}}(1-\text{SSIM})$ is the standard photometric loss, $\mathcal{L}_{\text{sky}}$ is Eq. 10, and $\mathcal{L}_{\text{reg}}$ includes:

- **Surfel regularization**: normal consistency and depth distortion losses from 2DGS [3]
- **Sun regularization**: penalty if ambient-to-sun ratio exceeds 0.3, L2 on color correction toward unity, L2 on sky SH to prevent degenerate solutions

TABLE I: Novel-view synthesis quality on NeRF-OSR scenes.

Method	PSNR ↑	SSIM ↑	LPIPS ↓
NeRF-OSR [5]	[TODO:]	[TODO:]	[TODO:]
LumiGauss [4]	[TODO:]	[TODO:]	[TODO:]
SunGauss (Lambert)	[TODO:]	[TODO:]	[TODO:]
SunGauss (PBR)	[TODO:]	[TODO:]	[TODO:]

V. Day-Cycle Relighting

Our explicit sun decomposition enables relighting by simply changing the sun direction $\mathbf{l}$ and elevation θ. We provide a day-cycle renderer that:

1) Computes a solar trajectory from geographic coordinates (latitude, day-of-year) using standard solar position equations [12]—declination $\delta = 23.45° \sin(360°(284 + d)/365)$, hour angle, and azimuth/elevation conversion
2) For each sun position, evaluates the sun color prior (Eq. 2), computes N·L shading with the new direction, renders a shadow map, and composites the final image
3) Produces a video sequence showing realistic time-of-day illumination changes with moving shadows

Since the sky SH, albedo, and geometry are shared across all images, only the direct term and shadow map change—ambient and sky contributions can be held fixed or adapted to the new sun elevation.

VI. Experiments

[TODO: This section to be filled with experimental results.]

A. Datasets

We evaluate on the NeRF-OSR [5] benchmark, which provides outdoor scenes captured under varying illumination with COLMAP reconstructions. We use the `lk2` and `st` scenes with associated sun direction metadata and sky masks.

B. Implementation Details

We train for 40,000 iterations with warmup $T_w = 20{,}000$ and shadow start $T_s = 20{,}500$. Shadow maps use 8192×8192 resolution with bias $b = 0.1$. The sun model parameters are optimized with learning rate 5×10^{-3} for intensities and 1×10^{-3} for sky SH. Albedo learning rate is 2.5×10^{-3}. When adaptive densification is enabled, we use grid resolution $G = 48$, interval $T = 250$, and budget $B = 4096$.

C. Baselines

- **LumiGauss** [4]: Per-image SH environment maps with 2DGS. No explicit sun model or shadow computation.
- **NeRF-OSR** [5]: NeRF-based outdoor relighting with SH decomposition.
- **[TODO: Additional baselines as needed.]**

D. Novel-View Synthesis

E. Relighting Quality

[TODO: Evaluate relighting with held-out illumination conditions. Show day-cycle results qualitatively. Compare shadow quality against baseline (implicit SH shadows).]

TABLE II: Ablation study on the `lk2` scene.

Configuration	PSNR ↑	SSIM ↑	LPIPS ↓
Full model	**[TODO:]**	**[TODO:]**	**[TODO:]**
w/o shadows	**[TODO:]**	**[TODO:]**	**[TODO:]**
w/o sun color prior	**[TODO:]**	**[TODO:]**	**[TODO:]**
w/o sky mask	**[TODO:]**	**[TODO:]**	**[TODO:]**
w/o adaptive dens.	**[TODO:]**	**[TODO:]**	**[TODO:]**
w/o global sky SH	**[TODO:]**	**[TODO:]**	**[TODO:]**

F. Ablation Studies

VII. Limitations and Future Work

a) Single directional light.: Our model assumes a single dominant sun direction. Overcast conditions with diffuse illumination are handled only through the ambient term, which may be insufficient for accurate relighting under heavily clouded skies.

b) Shadow map resolution.: The discrete shadow map introduces aliasing at object boundaries. Higher resolutions mitigate this at increased memory cost. Percentage-closer filtering or variance shadow maps could improve edge quality.

c) Static geometry.: Like the underlying Gaussian splatting, our method assumes a static scene. Transient objects (pedestrians, vehicles) present in the training images can cause artifacts and are not explicitly handled.

d) Normal quality.: Lambert shading quality depends on the accuracy of surfel normals. In regions with poor multi-view coverage, normals may be noisy, leading to incorrect N·L values and shading artifacts.

e) Future directions.: Promising extensions include learnable shadow bias, cascaded shadow maps for multi-scale scenes, differentiable shadow computation for end-to-end optimization, and integration with sky models for full environment relighting including cloud rendering.

VIII. Conclusion

We presented SunGauss, a method for physically-based outdoor scene relighting using Gaussian splatting. By replacing unconstrained per-image SH environments with an explicit directional sun model, we enable geometry-based shadow computation, atmospheric color priors, and realistic day-cycle relighting—capabilities that are fundamentally out of reach for per-image latent representations. Our adaptive densification and learnable sky distinction further improve reconstruction quality. **[TODO: Summary of quantitative improvements.]**

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